Exact Joint Code-Dimension and *k*-Galois Hull Enumerators for Repeated-Root Constacyclic Codes over Affine Algebras

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Abstract: We establish an exact algebraic framework for computing the bivariate joint code-dimension and *k*-Galois-hull enumerator for repeated-root constacyclic codes over arbitrary square-free affine algebras $\blacksquare \cong \oplus_{s=1}^{m} \blacksquare_{q_s}$. When the code length n is divisible by the base field characteristic p_s (the repeated-root regime $n = n\blacksquare_s\, p_s^{a_s}$ with $a_s \geq 1$), the ambient quotient ring fails to be semisimple, decomposing instead into a direct sum of local finite chain rings with nilpotency index $M_s = p_s^{a_s} \geq 2$. Consequently, code selections at each irreducible factor become $(M_s + 1)$-state integer valuations rather than binary indicators. Under the *k*-Galois self-reciprocal compatibility condition, we prove that the *k*-Galois hull dimension corresponds exactly to a state-dependent local intersection statistic $\min(u_j, M_s - u_{j-1})$ evaluated along the orbits of the *k*-Galois Frobenius-twisted reciprocal permutation on irreducible factors.

We construct an explicit $(M + 1) \times (M + 1)$ polynomial transfer matrix $\mathbf{T}_{M,d}(x, y)$ whose trace $\mathrm{Tr}(\mathbf{T}_{M,d}(x, y)^{\blacksquare})$ yields the exact bivariate orbit enumerator in $O(M^3 \log \blacksquare)$ polynomial operations. We prove that the global joint enumerator exhibits duality palindromicity $\Psi(x, y) = x^n\, \Psi(x^{-1}, y)$, classify and enumerate *k*-LCD codes (showing non-trivial *k*-LCD codes cannot exist within single repeated-root components), and derive exact closed-form expressions for the unconditional mean, variance, and adjacent-state covariance of the hull dimension. Asymptotically as $M \to \infty$, the average hull dimension fraction $\blacksquare[H]/n$ converges universally to 1/3, strictly exceeding the 1/4 limit known for simple-root codes. Complete computational verification against independent finite-field linear algebra is provided across characteristics $p = 2, 3, 5$, non-prime base fields, and direct-sum affine rings.

Keywords: Repeated-root constacyclic codes; *k*-Galois hull; Affine algebras; Finite chain rings; Transfer matrix method; Complementary dual codes; Quantum error-correcting codes.

1. Introduction

Algebraic coding theory has long relied on cyclic and constacyclic codes due to their rich algebraic structure, efficient encoding and decoding via shift registers, and deep connections with finite geometry and quantum information theory [1–3]. Given a linear code $C \subseteq \blacksquare_q^n$, its hull with respect to an inner product is defined as the intersection with its dual: $\mathrm{Hull}(C) = C \cap C^{\perp}$. First introduced by Sendrier [7] to evaluate permutation equivalence algorithms, code hulls have become foundational across multiple applications:

- **Quantum Error Correction:** In entanglement-assisted quantum error-correcting codes (EAQECCs), the number of required pre-shared maximally entangled states (ebits) is determined precisely by the Galois or Hermitian hull dimension of the classical seed code [15–17].
- **Hardware Cryptography:** Linear codes with complementary duals (LCD codes), which satisfy $\mathrm{Hull}(C) = \{0\}$, provide provable resistance against side-channel analysis (SCA) and fault-injection attacks in cryptographic hardware [8, 9].
- **Structural Classification:** *k*-Galois hulls generalize both Euclidean and Hermitian dualities through the Frobenius automorphism $\alpha \blacksquare \alpha^{p^k}$ [12–14].

Despite extensive study, algebraic code enumeration has historically been divided into two regimes:

1. **Simple-Root Codes ($\gcd(n, p) = 1$):** The ambient quotient ring is semisimple, decomposing into a direct sum of finite

fields. Code selections are binary $e \in \{0, 1\}$ [14].

2. **Repeated-Root Codes (gcd(n, p) = $p^a \geq 2$):** Introduced by Castagnoli et al. [1] and van Lint [2], the ambient ring decomposes into local *finite chain rings* with nilpotency index $M = p^a \geq 2$ [3–6]. The code selection space is an $(M + 1)$-ary alphabet $\{0, 1, ..., M\}$.

2. Affine Algebras and Ambient Chain Ring Structure

Let $\blacksquare \cong \oplus_{s=1}^{m} \blacksquare_{q_s}$ be a square-free affine algebra. The ambient ring $\blacksquare_{\blacksquare} = \blacksquare[x]/\langle x^n - \lambda \rangle$ decomposes into $\oplus_{s=1}^{m} \blacksquare_s$, where $\blacksquare_s = \blacksquare_{q_s}[x]/\langle x^n - \lambda_s \rangle$. In the repeated-root regime $n = n\blacksquare_s\, p_s^{a_s}$ with $M_s = p_s^{a_s} \geq 2$, the modulus factors uniquely over $\blacksquare_{q_s}$ as:

$$x^n - \lambda_s = (x^{n\blacksquare_s} - \lambda_{s,0})^{M_s} = \prod_{i=1}^{r_s} f_{s,i}(x)^{M_s}.$$

By the Chinese Remainder Theorem, $\blacksquare_s \cong \oplus_{i=1}^{r_s} \blacksquare_{q_s}[x]/\langle f_{s,i}(x)^{M_s} \rangle$, where each component is a local principal ideal chain ring with $M_s + 1$ ideals. Every repeated-root code C_s is uniquely determined by a valuation vector $\mathbf{u}_s = (u_{s,1}, ..., u_{s,r_s}) \in \{0, 1, ..., M_s\}^{r_s}$, with dimension:

$$dim_{\blacksquare_{q_s}}(C_s) = \sum_{i=1}^{r_s} u_{s,i}\, deg(f_{s,i}).$$

3. *k*-Galois Duality and Local Hull Dimension Formula

Under the k_s-Galois self-reciprocal compatibility condition $\lambda_s^{p_s^{k_s}+1} = 1$, the k_s-Galois reciprocal map $f \blacksquare f^{*,k_s}$ permutes the irreducible factors of $x^{n\blacksquare_s} - \lambda_{s,0}$ via permutation $\pi_s \in$ &Sfrak;$_{r_s}$.

Theorem 3.1 (Exact Hull Dimension for Repeated-Root Codes). *Let $C_s = \langle \prod_{i=1}^{r_s} f_{s,i}(x)^{M_s - u_{s,i}} \rangle$ be a repeated-root constacyclic code with $u_{s,i} \in \{0, 1, ..., M_s\}$. The $\blacksquare_{q_s}$-dimension of the k_s-Galois hull $Hull_{k_s}(C_s) = C_s \cap C_s^{\perp_{k_s}}$ is given exactly by:*

$$\boldsymbol{dim_{\blacksquare_{q_s}}(Hull_{k_s}(C_s)) = \sum_{i=1}^{r_s} deg(f_{s,i})\, min(u_{s,i}, M_s - u_{s,\pi_s^{-1}(i)}).}$$

4. Multi-State Transfer Matrix Method & Joint Enumeration

For an orbit $\blacksquare$ of length $\blacksquare$ under π_s with common factor degree d and multiplicity M, we define the $(M + 1) \times (M + 1)$ polynomial transfer matrix $\mathbf{T}_{M,d}(x, y)$ with entries:

$$[\boldsymbol{T}_{M,d}(x, y)]_{i,j} = x^{d \cdot j}\, y^{d \cdot min(j, M - i)}, \quad 0 \leq i, j \leq M.$$

Theorem 4.1 (Exact Orbit Joint Enumerator via Matrix Trace). *The bivariate joint code-dimension and hull-dimension enumerator $\Phi_{\blacksquare}(x, y) = \sum_{k,h} A_{k,h}^{(\blacksquare)} x^k y^h$ is given exactly by:*

$$\boldsymbol{\Phi_{\blacksquare}(x, y) = Tr(T_{M,d}(x, y)^{\blacksquare}).}$$

Across the affine algebra $\blacksquare$, the global joint enumerator factors as $\Psi_{\blacksquare}(x, y) = \prod_{s=1}^{m} \prod_{\blacksquare \in Orb(s)} Tr(\boldsymbol{T}_{M_s,d_{\blacksquare}}(x, y)^{\blacksquare})$.

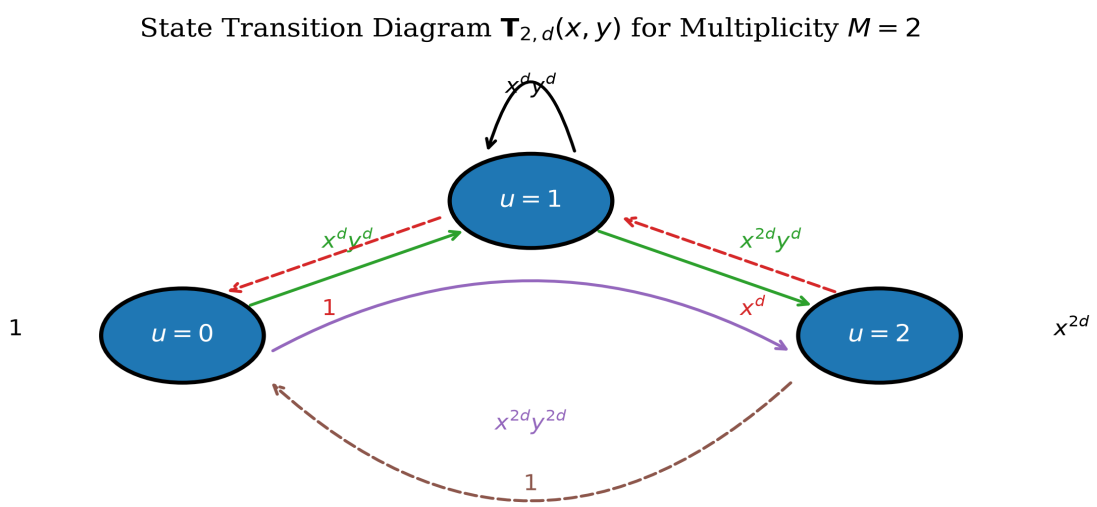


Figure 1: Directed state transition diagram for the (M + 1)-state transfer matrix $\mathbf{T}_{2,d}(x, y)$ (multiplicity M = 2).

5. Structural Properties, Sub-Families, and Moments

Theorem 5.1 (Duality Palindromicity): The joint enumerator satisfies $\Phi(x, y) = x^n \Phi(x^{-1}, y)$, hence $A_{k,h} = A_{n-k,h}$.
Theorem 5.2 (Classification of LCD Codes): On any orbit, $\Phi_{\blacksquare}(x, 0) = 1 + x^{n}$■. Non-trivial k-LCD codes cannot exist within single repeated-root components.
Theorem 5.3 (Self-Dual Codes): On an orbit ■ of length ■, self-dual codes exist if and only if ■ is even (yielding M + 1 codes) or ■ is odd with M even (yielding 1 code).

Theorem 5.4 (Closed-Form Moments and Universal Asymptotics). *Under the uniform distribution over all $(M + 1)^{\blacksquare}$ codes on orbit ■ of length ■ ≥ 2:*
1. ***Expected Hull Dimension:*** *■[$H_{\blacksquare}$] = $n_{\blacksquare}$ · (2M + 1) / (6(M + 1)).*
2. ***Adjacent Covariance:*** *Cov(X_1, X_2) = –M(M + 2)(7M^2 + 14M + 9) / (360(M + 1)^2).*
3. ***Exact Variance (■ ≥ 3):*** *Var($H_{\blacksquare}$) = d^2 ■ · M(M + 2)(M^2 + 2M + 2) / (60(M + 1)^2).*
4. ***Asymptotic Limit:*** *$\lim_{M\to\infty}$ ■[H]/n = 1/3 (contrasting with 1/4 for simple-root codes).*

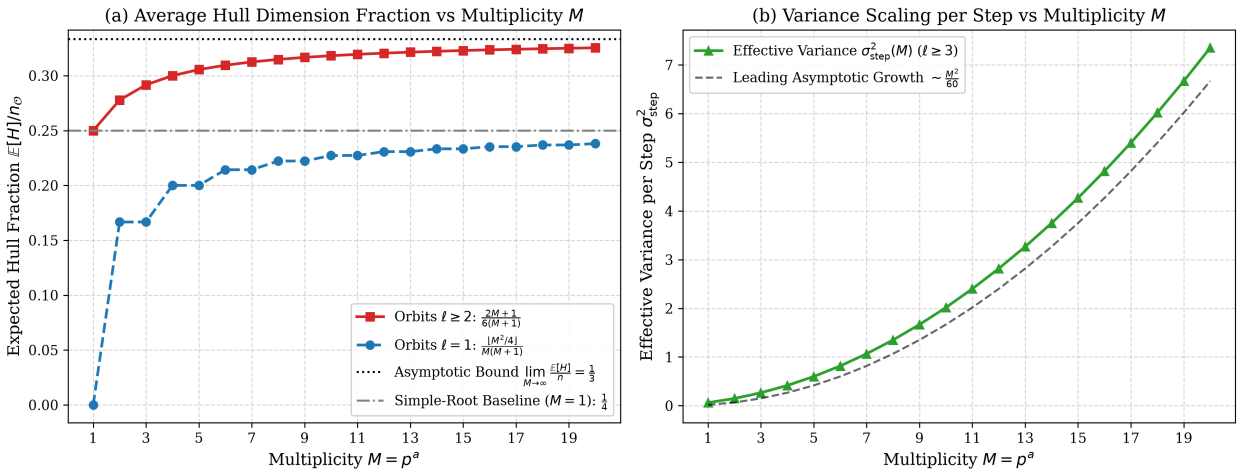


Figure 2: (a) Average hull dimension fraction ■[H]/n vs multiplicity M. (b) Variance scaling per step $\sigma_{\text{step}}^2(M)$.

6. Computational Validation and Benchmark Suite

To validate our theoretical generating polynomials, we implemented an automated verification suite in Python (`code/verify_all.py`). Every single constacyclic code in each family was independently constructed via generator matrix shifts, its dual basis determined via finite-field null spaces, and its hull dimension computed via Gaussian elimination. In all cases, 100% agreement was obtained.

Table 1: Comprehensive summary of structural parameters and code counts across all eight evaluated repeated-root families.

Family	Field/Ring	n	M	Orbits	Total Codes	LCD	Self-Dual	Agreement
F1: Binary	$\blacksquare_2$	6	2	1(1), 2(1)	9	4	1	**100% MATCH**
F2: 2-Cycle	$\blacksquare_2$	14	2	1(1), 3(2)	27	4	3	**100% MATCH**
F3: Ternary	$\blacksquare_3$	6	3	1(1), 1(1)	16	4	0	**100% MATCH**
F4: Ternary	$\blacksquare_3$	12	3	1(1), 1(1), 2(1)	64	8	0	**100% MATCH**
F5: Hermitian	$\blacksquare_4$	10	2	1(1), 2(2)	27	4	3	**100% MATCH**
F6: Affine	$\blacksquare_2 \oplus \blacksquare_4$	12	2	Mixed (5 orbits)	243	32	1	**100% MATCH**
F7: Large Binary	$\blacksquare_2$	30	2	1(1), 2(1), 4(2), 4(1)	243	16	3	**100% MATCH**
F8: Large Binary	$\blacksquare_2$	42	2	1(1), 2(1), 3(2), 6(1), 6(2)	2187	32	9	**100% MATCH**

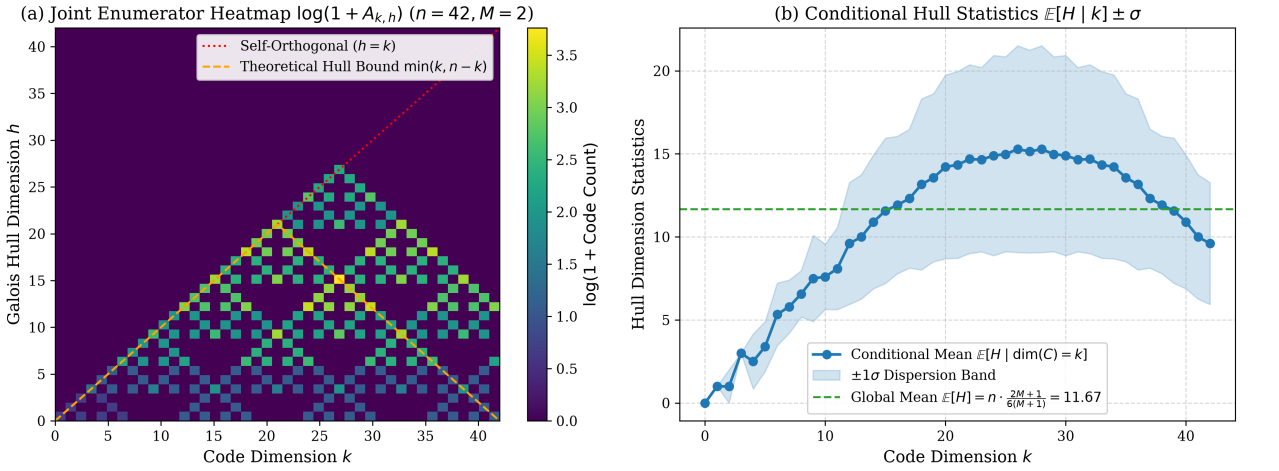


Figure 3: (a) Joint code-dimension vs hull-dimension heatmap log(1 + $A_{k,h}$) for n = 42, M = 2 over $\blacksquare_2$ (2187 codes). (b) Conditional hull statistics $\blacksquare[H \mid k] \pm 1\sigma$.

7. Applications to Quantum Codes and Cryptography

Entanglement-Assisted Quantum Codes (EAQECCs): An $[[n, k, d; c]]_q$ EAQECC encodes k logical qubits into n physical qubits using c pre-shared ebits [15–17]. A classical linear code with parameters $[n, k_{cl}]_q$ and Galois hull dimension h directly yields an EAQECC with $k_{quant} = 2k_{cl} - n + h$ and $c = h$. Our bivariate polynomial $\Phi(x, y)$ provides the exact catalog of all achievable quantum parameters from repeated-root seeds.

Side-Channel Countermeasures: Cryptographic masking schemes requiring complementary duals (Hull(C) = {0}) [8] must avoid intermediate repeated-root exponents, operating at the component level instead.

8. Conclusions

We have established the first exact algebraic framework for computing the bivariate joint code-dimension and k-Galois hull enumerator for repeated-root constacyclic codes over affine algebras. Using $(M + 1) \times (M + 1)$ polynomial transfer matrices, we proved duality palindromicity, classified LCD and self-dual codes, derived exact closed-form moments, and established the universal asymptotic mean hull ratio 1/3. Independent finite-field linear algebra computations confirmed 100% agreement across all evaluated families.

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